enormous speed, one-sixth that of light, meant that 3C273 was 2 billion light-years from Earth, nearly the farthest object that had ever been seen at that time. From its brightness and its distance, Schmidt concluded that 3C273 puts out 4 trillion times more power in light than the Sun, and a hundred times more power than the brightest galaxies!

This prodigious power fluctuated on times as short as a month, so most of the light must be coming from an object so small that the light can travel across it in one month's time—far smaller than the distance from Earth to the nearest star, Proxima Centauri. And other quasars with almost as much power fluctuated on times of a few hours, so they had to be not much larger than our solar system. *One hundred times the power of a bright galaxy, coming from a region the size of our solar system; that was phenomenal!*

Black Holes and Accretion Disks

How could so much power come out of a region so small? When we think about the fundamental forces in Nature, there are three possibilities: chemical energy, nuclear energy, or gravitational energy.

Chemical energy is the energy released when molecules combine together to make new kinds of molecules. An example is burning gasoline, which combines oxygen from the air with gasoline molecules to make water and carbon dioxide, and a lot of heat. The power from that would be far, far, far too little though.

Nuclear energy results when atomic nuclei combine together to make new atomic nuclei. Examples are an atomic bomb, a hydrogen bomb, and the burning of nuclear fuel inside a star. Though this can be far more powerful than chemical energy (think of the difference between a gasoline fire and a nuclear bomb), astrophysicists couldn't